

Reinforcement Learning Basics

Reinforcement learning is a branch of machine learning in which an agent learns to make decisions by interacting with an environment and receiving feedback in the form of rewards or penalties. Unlike supervised learning, reinforcement learning does not rely on labeled examples; instead, the agent must discover which actions yield the best long-term outcomes through trial and error. The core components of a reinforcement learning system include the agent, the environment, the set of possible actions, and a reward signal that guides learning. At each time step, the agent observes the current state of the environment, selects an action based on its current policy, and receives a reward along with a new state. The goal of the agent is to learn a policy that maximizes cumulative reward over time. A key challenge in reinforcement learning is balancing exploration, trying new actions to discover their effects, with exploitation, choosing actions already known to yield good rewards. Popular algorithms include Q-learning, which estimates the value of taking a particular action in a particular state, and policy gradient methods, which directly optimize the agent's decision-making policy. Deep reinforcement learning combines these ideas with deep neural networks, enabling agents to handle high-dimensional inputs such as raw pixel data from video games. Notable achievements in the field include AI systems that have mastered complex games such as Go and chess at superhuman levels, as well as applications in robotics, where agents learn to perform physical tasks such as grasping objects or walking through simulated and real-world trial and error.